

1. Write the full form and one-line use of: HTTP, HTTPS, FTP, SMTP.

Term	Full Form	Use
HTTP	HyperText Transfer Protocol	Used for transferring web pages between a browser and a web server.
HTTPS	HyperText Transfer Protocol Secure	Used for secure communication between a browser and a web server.
FTP	File Transfer Protocol	Used for transferring files between computers over a network.
SMTP	Simple Mail Transfer Protocol	Used for sending emails over the Internet.

2. What is the difference between IPv4 and IPv6? (2 lines)

- **IPv4** uses 32-bit addresses and supports about 4.3 billion unique addresses.
 - **IPv6** uses 128-bit addresses and supports a much larger number of unique addresses with improved security and efficiency.
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3. Run nslookup google.com on your system and write down the IP address you get.

I cannot run commands on your computer. You can run:

```
nslookup google.com
```

Example output (may vary by location and time):

```
google.com
```

```
Address: 142.250.193.14
```

IP Address Obtained: 142.250.193.14 (*example only; your result may be different*).

4. What do these HTTP status codes mean: 200, 301, 404, 500, 403?

Status Code Meaning

Status Code Meaning

200	OK – The request was successful.
301	Moved Permanently – The resource has been moved to a new URL.
404	Not Found – The requested page or resource does not exist.
500	Internal Server Error – The server encountered an unexpected problem.
403	Forbidden – Access to the requested resource is denied.

5. Explain DNS hierarchy (root, TLD, domain, subdomain) with www.google.com as an example.

DNS Level	Example	Description
Root	. (root)	The highest level in the DNS hierarchy.
TLD (Top-Level Domain)	.com	Identifies the category of the domain.
Domain	google.com	The registered website name.
Subdomain	www.google.com	A subdivision of the main domain used to organize services.